Substance Harm Across New Zealand

Purpose and Research Question

Substance harm can affect individuals, communities and the wider public through health, social and economic impacts. Our research investigated how substance-harm-related police activity differs across New Zealand, with a particular focus on police districts.

Our main questions were whether police districts differ from one another, whether some districts have higher police action rates than others, and whether the types of drugs involved also vary across districts. This was a change from our earlier focus on simply comparing the North and South Islands. Looking more closely at police districts allowed us to identify local differences that could be hidden by a broad island-level comparison. This reflects the focus used in our final presentation.

Our intended audience is the New Zealand Government. The findings may be useful when considering where substance-harm prevention, treatment and community support may be most needed.

Data and Approach

Our analysis used information from NZ Police, the New Zealand Health Survey and Stats NZ population estimates.

The main NZ Police dataset contained police actions relating to drugs from January 2016 to June 2023. Police actions were used as our main outcome. However, it is important to recognise that police actions represent policing activity and police responses to offences. They do not directly measure the true level of drug use or drug-related crime in a community. This distinction was an important ethical consideration throughout our analysis.

Population size also differs considerably between police districts. Comparing only the number of police actions would therefore be misleading because larger areas could naturally have more actions simply because more people live there.

To address this, we converted police actions to rates per 10,000 people using population estimates. This made comparisons between districts more meaningful and allowed areas with different population sizes to be considered on a similar scale.

Differences Between Police Districts

One of the clearest findings was that overall police action rates varied substantially between police districts. The Bay of Plenty had the highest overall police action rate at approximately 45.8 actions per 10,000 people. Tasman was next at approximately 40.0, followed closely by

Eastern at 39.5. At the other end of the comparison, Auckland had the lowest rate at approximately 20.8 actions per 10,000 people.

This means police action rates were not uniform across New Zealand. The difference between the Bay of Plenty and Auckland was substantial, with the Bay of Plenty's rate more than twice Auckland's.

However, this result should not be interpreted as showing that people in Bay of Plenty necessarily use more drugs than people in Auckland. Police action rates may also be influenced by policing practices, local priorities, enforcement activity and other factors.

The important finding for government is therefore that district-level variation exists and is large enough that national or island-level comparisons alone may hide important local differences.

Districts Also Differ by Drug Type

We then examined the police-action data for particular drug types, focusing on cannabis and methamphetamine.

The results showed a different picture from the overall police-action comparison. Among the districts displayed in our analysis, Tasman had the highest rates for both cannabis and methamphetamine, while Auckland had the lowest rates for both. For example, Tasman's cannabis rate was 101.5 per 10,000 people, compared with 46.4 in Bay of Plenty and 16.5 in Auckland. For methamphetamine, Tasman's rate was 28.0 per 10,000, compared with 23.3 in Bay of Plenty and 10.0 in Auckland. Crime Busters

This finding is important because Bay of Plenty had the highest overall police action rate, while Tasman had the highest rate for these two particular drug types among the districts shown. This demonstrates that a district can have a high overall police action rate without having the highest rate for every individual drug type. The districts therefore differ not only in the amount of police activity but also in their drug mix.

For policy decisions, this suggests that different districts may require different responses rather than one national approach to substance harm.

Police Responses Also Differ Between Districts

Our analysis also suggested that the police response to the same drug was not identical across districts. For cannabis, the proportion shown in our presentation ranged from 47.6% in Canterbury to 69.5% in Southern. For methamphetamine, it ranged from 12.2% in Canterbury to 34.0% in Southern. Our presentation also showed an increase in alternative action across all drug actions following the 2019 amendment. Crime Busters

This reinforces the importance of being careful when interpreting police data. Differences between districts may reflect differences in substance-related activity, but they may also reflect differences in how offences are dealt with.

Misuse of Drugs Amendment Act 2019

The Misuse of Drugs Amendment Act (2019) was passed on the 7th of August 2019 and came into effect on the 13th of August 2019.

The Act amends the Misuse of Drugs Act (1975).

The Principal act that regulates and controls the classification, production, supply, possession and use of drugs.

The 2019 amendment made 3 key changes:

- Classified the synthetic cannabinoids AMB-FUBINACA and 5F-ADB as class A* drugs.
    - On the advice of an Expert Advisory Committee
- Affirms the existing Police discretion to prosecute for possession and use of controlled drugs.
    - The Act specifies that consideration should be given to health-based approaches when prosecuting in the public interest.
- Enables temporary drug class orders to be issued by the Minister of Health.
    - The order immediately classifies a substance for a temporary period. Where the substance is to be treated as a Class C** controlled drug (including penalties).

*drugs that pose a very high risk of harm

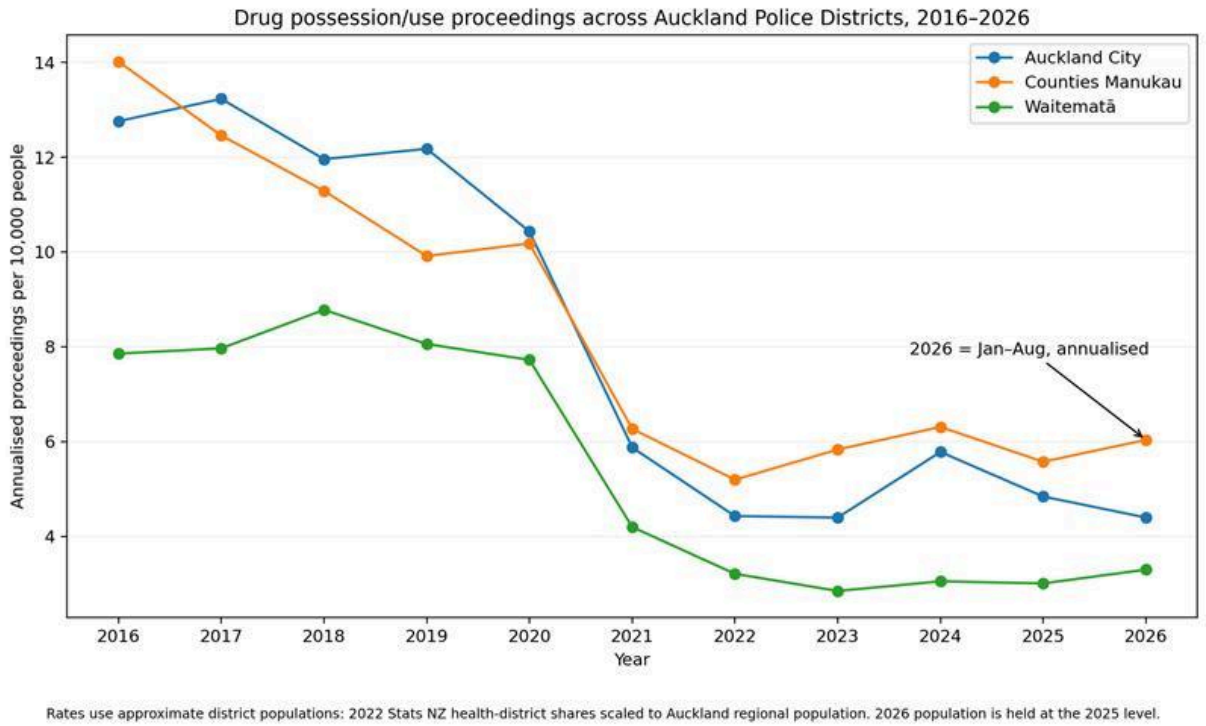
**drugs that pose a moderate risk of harm

Auckland as an Important Case

To examine Auckland in greater detail, we separated the region into Auckland City, Counties Manukau and Waitematā Police Districts. Because the original police-actions dataset ended in June 2023, we used the second NZ Police proceedings dataset to examine whether the pattern continued through 2026. To maintain a consistent measure over time, we restricted this analysis to possession/use proceedings.

The results show that the decline observed after 2020 was not driven by a single Auckland district. Between 2020 and 2022, possession/use proceeding rates fell by approximately 58% in Auckland City, 49% in Counties Manukau and 58% in Waitematā. Rates remained well below their pre-2021 levels through 2026. Waitematā consistently had the lowest rate, while Counties Manukau showed a modest recovery after 2022 and had the highest rate of the three districts by 2026.

This suggests that Auckland's overall decline reflects a broad regional pattern rather than one unusually low district. However, meaningful differences remain within Auckland itself, showing that treating Auckland as a single region may hide important district-level variation.



These district-level population rates are approximate because the available population boundaries do not perfectly match Police District boundaries. In addition, the proceedings dataset measures a different police outcome from the original police-actions dataset, so the two datasets are used as complementary evidence rather than treated as directly interchangeable.

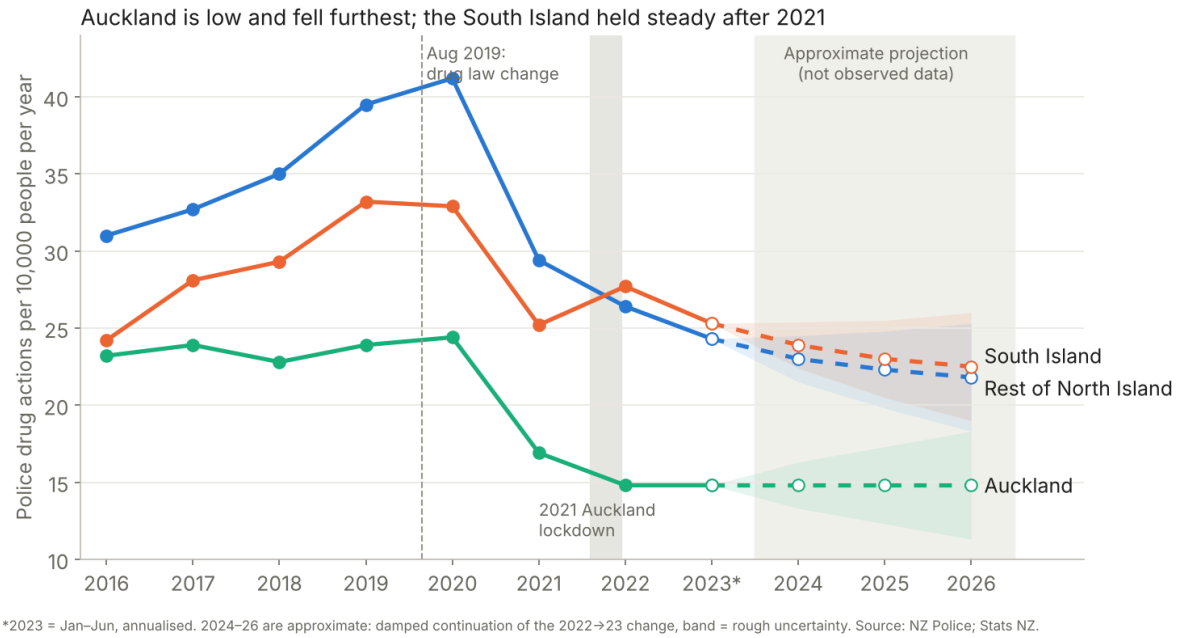
2024–2026 Projection

An extended three-group graph was also produced to explore what the existing pattern could look like through 2026.

2024–26 are approximate projections, not real data. The police data we have ends in June 2023, and we couldn't find any public update. The dashed lines continue the 2022→23 trend, slowing each year, and the shaded band indicates the approximate uncertainty.

What it shows:

- Auckland stays low at about 15 per 10,000 after its 2021 fall.
- South Island and Rest of North Island drift down slowly to about 22–23 per 10,000 by 2026, with the South staying slightly higher.
- The 2023 point is only Jan–Jun, annualised, so the starting point is a bit shaky.



The main finding hasn't changed: the South Island's higher rate from 2022 is mostly because Auckland's rate fell, and this is about policing, not drug use.

Effect of Police Methods on Drug Use

Police methods can influence drug use in New Zealand through enforcement strategies such as arrests, drug seizures, and prosecution, as well as health-based approaches such as treatment referrals and prevention programmes. Stricter enforcement may discourage some individuals from using drugs and disrupt the supply of illegal substances. However, continued drug consumption in New Zealand suggests that enforcement alone may not be sufficient to reduce overall drug use.

A health-based approach that focuses on treatment, education, and harm reduction may also help address the underlying causes of problematic drug use. It is important to recognise that the relationship between policing and drug use can work in both directions, as increased drug use may also lead to increased police activity. Therefore, further research is needed to determine whether police methods have a statistically significant effect on drug use in New Zealand and whether certain approaches are more effective than others.

Ethical Considerations

Several ethical issues are important when interpreting our findings. First, police actions measure policing, not drug use or crime. A higher police action rate could partly result from differences in enforcement, police priorities or how offences are dealt with.

For this reason, our results should not be described as showing that one district's population uses more drugs than another. We also avoided drawing conclusions based on ethnicity

because the available data did not provide sufficient evidence to disentangle ethnicity from other potential influences.

More generally, this is observational data. We can describe geographic and temporal associations, but we should be cautious about claiming that events such as COVID-19 or the 2019 law change directly caused changes in police action rates.

Limitations

There are several limitations to the analysis. The original NZ Police actions dataset ends in **2023**, meaning that later trends cannot be directly observed using the same outcome. A second NZ Police proceedings dataset was used for some later analysis, but it measures a different police outcome and should therefore be treated as complementary rather than directly interchangeable with the first dataset.

Police district boundaries also only approximately match the geographic boundaries available for population estimates. This means the district-level rates are approximate rather than exact.

The 2019 law amendment and COVID-19 period during 2020–2021 both occurred during our study period and may have disrupted the trend. Finally, police data alone cannot provide a complete picture of substance harm.

Conclusion

Our analysis shows that substance-harm-related police activity varies across New Zealand, and that examining individual police districts provides a clearer picture than relying solely on a North–South Island comparison. Bay of Plenty had the highest overall police-action rate, while Auckland had the lowest. However, when cannabis and methamphetamine were considered separately, Tasman had the highest rates among the districts shown.

The Auckland analysis also showed that its lower rate was not caused by one single district, because Auckland City, Counties Manukau and Waitematā all showed substantial declines after 2020. At the same time, the 2019 drug law change and the COVID-19 period occurred during our study period, so these trends should be interpreted carefully. Police-action data reflects policing activity and decision-making rather than actual drug use.

Overall, the findings suggest that substance-harm responses should consider local district-level differences and should use police data alongside health and wastewater evidence. This would give the government a more complete picture of where support, treatment and prevention services may be most needed.

Recommendations

Responses to substance harm should consider differences between individual police districts rather than relying only on national averages or North–South Island comparisons.

Police-action data should also be considered alongside wastewater and health evidence to provide a more complete picture of local substance harm.

Where several sources indicate higher substance harm or gaps in support, additional treatment and community support could then be directed to those areas.

Police, health services and community organisations should also work together to connect people experiencing substance harm with appropriate early support and treatment.

Technical supplement

Analytic Steps:

- We collected official data from NZ Police, the New Zealand Health Survey, and Stats NZ population estimates.
- Used Police Actions as the main outcome and removed the duplicated “Running Sum of Police Actions” column.
- Matched police districts approximately to regions and linked them with Stats NZ population estimates.
- Converted Police Actions into rates per 10,000 people so areas with different population sizes could be compared fairly.
- Compared the North Island and South Island, including a separate comparison of the North Island without Auckland.
- Split the analysis into three time periods: Jan 2016 – Aug 2019, Sep 2019 – Dec 2021, and Jan 2022 – Jun 2023.
- Used a quasi-Poisson regression on monthly counts, with month effects and population included as an offset.
- Then we compared different drug types, including all drugs, cannabis, and methamphetamine.
- Checked seasonality to see whether both islands followed similar monthly patterns and whether there was an extra South Island summer effect.
- Compared individual police districts to see whether district-level differences were larger than the overall North–South difference.
- Calculated annual rates for Auckland, the Rest of the North Island, and the South Island to examine how the pattern changed over time.
- Used the later proceedings dataset to check whether the pattern continued beyond 2023.

- Interpreted results carefully because police actions measure policing activity, not actual drug use or crime.
- Noted that police district boundaries and Stats NZ population boundaries do not match perfectly, so some district-level rates are approximate.
- Treated the later proceedings dataset as supporting evidence only, because it measures a different police outcome from the original police-actions dataset.

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